

Accelerated Decentralized Gradient Descent for Strongly Convex Optimization

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Abstract. Decentralized stochastic optimization is a standard framework for large-scale learning over networks where a central coordinator is unavailable or undesirable. In this paper we propose *Multi-Gossip Accelerated DSGD* (MC-ADSGD), a decentralized stochastic optimization algorithm that combines a Nesterov-type primal–dual extrapolation with a multi-round accelerated gossip averaging primitive (FGA). At each outer iteration, the proposed algorithm performs two calls to FGA with gossip depth R , thereby allowing multiple rounds of communication within one optimization step. Under standard smoothness, strong convexity, and bounded-variance assumptions, we establish a non-asymptotic convergence guarantee that makes explicit the dependence on $(\mu, L, \sigma^2, n, R, \beta)$ and on a heterogeneity quantity G^* measuring the mismatch between local and global optima. The resulting bound consists of a linearly decaying transient term together with stochastic and heterogeneity-dependent residual terms. The corresponding communication-complexity consequence exhibits the scaling

$$\tilde{O}\left(\frac{\sigma^2}{\mu n \epsilon} + \sqrt{\frac{L}{\mu}} \frac{1}{\sqrt{1-\beta}}\right),$$

up to logarithmic factors, with the heterogeneity contribution entering through the residual floor in the convergence bound. This also yields the guideline $R = \tilde{O}(1/\sqrt{1-\beta})$ for the gossip depth.

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Key words: Decentralized optimization, stochastic gradient methods, distributed learning, communication complexity, gossip algorithms

1 Introduction

Decentralized stochastic optimization has become a standard tool for large-scale learning when data and computation are distributed across many workers but a central parameter server is either undesirable or unavailable. In such systems, each node only exchanges messages with its neighbors according to an underlying communication graph. This architecture improves robustness, alleviates the single-point bottleneck of centralized synchronization, and naturally matches settings where data are generated and stored at the

edge. At the same time, local computation and network communication are tightly coupled: aggressive local updates may accelerate descent, while insufficient mixing amplifies disagreement across the network and slows global convergence.

A large body of work has therefore focused on characterizing the trade-off between optimization progress and consensus error for decentralized stochastic gradient methods. Classical decentralized SGD (often called D-SGD or D-PSGD) uses one consensus step per iteration and achieves convergence guarantees that explicitly depend on the spectral gap of the mixing matrix, the condition number of the objective, and the variance of the stochastic gradients [5, 14, 17]. Subsequent methods such as EXTRA, exact diffusion, and gradient-tracking variants reduce steady-state bias and improve communication efficiency by better correcting the drift between local models and the network average [7, 11, 13, 18]. In parallel, accelerated methods and multi-consensus schemes have demonstrated that several rounds of communication per iteration, possibly combined with polynomial (Chebyshev-type) gossip, can substantially improve the dependence on the spectral gap [2, 4, 12].

Despite substantial progress on decentralized optimization, it remains useful to understand how accelerated optimization dynamics and accelerated communication primitives interact in the stochastic strongly convex regime. In particular, one would like to clarify how multiple communication rounds within each outer iteration affect the overall dependence on the condition number, the network spectral gap, the stochastic noise level, and the residual terms induced by data heterogeneity.

Motivated by this question, we propose *Multi-Gossip Accelerated DSGD* (MC-ADSGD), a decentralized stochastic optimization algorithm that combines a Nesterov-type primal-dual extrapolation with two calls to a fast gossip averaging operator (FGA) per outer iteration. The gossip depth R controls the number of communication rounds inside each FGA call. In the parameter regime considered in our analysis, we further couple the mini-batch size and the gossip depth through the choice $B = R$, so that increasing R improves communication accuracy while also reducing gradient variance.

Our focus in this paper is on presenting the proposed MC-ADSGD method and making its convergence and communication-complexity implications explicit in the strongly convex stochastic setting. Under standard smoothness, strong convexity, and bounded-variance assumptions, we derive a Lyapunov-based convergence bound consisting of a linearly decaying term together with stochastic and heterogeneity-dependent residual terms. This bound leads to a communication-complexity consequence of order

$$\tilde{\mathcal{O}}\left(\frac{\sigma^2}{\mu n \epsilon} + \sqrt{\frac{L}{\mu}} \frac{1}{\sqrt{1-\beta}}\right),$$

up to logarithmic factors, while the heterogeneity quantity G^* enters through the residual accuracy floor.

The purpose of the present paper is therefore to give a self-contained presentation of the proposed algorithm and its analysis, and to isolate the communication-complexity consequence of the current proof in a form suitable for further comparison and extension.

Contributions. The main contributions of this work are as follows.

- We propose *Multi-Gossip Accelerated DSGD* (MC-ADSGD), a decentralized stochastic optimization algorithm that combines a Nesterov-type primal–dual extrapolation with a multi-round accelerated gossip primitive (FGA), and we specify the parameter regime analyzed in this paper.
- We derive an explicit strongly convex convergence guarantee for the proposed MC-ADSGD algorithm, whose dependence on $(\mu, L, \sigma^2, n, R, \beta)$ and on the heterogeneity quantity G^* is stated in closed form.
- We isolate the corresponding communication-complexity consequence, which exhibits a stochastic term of order $\sigma^2 / (\mu n \epsilon)$ and a network term of order $\sqrt{L/\mu} / \sqrt{1-\beta}$, up to logarithmic factors, together with the scaling guideline $R = \tilde{O}(1/\sqrt{1-\beta})$.
- We collect the technical derivations in the appendix and organize the proof through a coupled Lyapunov argument that tracks both network disagreement and optimization error.

2 Related Work

This section briefly reviews the most relevant literature on decentralized stochastic optimization, multi-round and accelerated gossip, and communication-efficient accelerated methods, and situates *Multi-Gossip Accelerated DSGD* within this landscape.

Decentralized stochastic gradient methods. Early decentralized stochastic gradient methods such as D-SGD and D-PSGD [5, 14] apply one consensus (or gossip) step per iteration and establish convergence guarantees that explicitly depend on the spectral gap of the underlying mixing matrix. These methods are simple and widely used, but their communication complexity can be suboptimal on poorly connected graphs, and their steady-state error typically reflects a nonvanishing disagreement component. Variants based on EXTRA and exact diffusion [13, 17, 18] as well as gradient-tracking schemes such as DIGing, NEXT, and related algorithms [7, 11, 16] mitigate this issue by tracking the global gradient more accurately and correcting local drift. Many of these works operate either in deterministic or diminishing-stepsize regimes; when stochastic gradients and constant stepsizes are considered, the resulting steady-state bounds often make the dependence on noise and network parameters only partially explicit.

Gradient tracking and exact diffusion. Gradient-tracking methods maintain auxiliary variables that estimate the average gradient across the network and use them to drive local updates toward the global optimum. In the deterministic setting, such methods achieve exact convergence and, in some cases, are known to be optimal up to logarithmic

factors in terms of the dependence on the condition number and the spectral gap [12]. In the stochastic strongly convex case, recent works have extended gradient tracking and exact diffusion to handle gradient noise and non-identical data distributions across nodes [10, 15]. Compared with these approaches, the proposed MC-ADSGD algorithm does not maintain an explicit gradient-tracking variable; instead, it combines a Nesterov-type primal–dual extrapolation with multi-round accelerated gossip and ties the batch size to the gossip depth. Our analysis focuses on making the communication complexity in this specific design explicit, rather than deriving a general-purpose optimality result for all gradient-tracking variants.

Multi-round and accelerated gossip. The idea of performing multiple gossip rounds per iteration to accelerate consensus dates back to early work on multi-consensus methods and polynomial filtering of the mixing matrix [2, 9]. Chebyshev-accelerated gossip schemes design a polynomial in the mixing matrix that minimizes the worst-case contraction factor over the spectrum, yielding a number of rounds that scales as $\tilde{O}(1/\sqrt{1-\beta})$ with the inverse square root of the spectral gap. Such primitives have been used both as stand-alone averaging mechanisms [2] and as building blocks within decentralized optimization algorithms [4, 12]. The FGA operator used in the proposed MC-ADSGD method is an instance of this general idea: it implements an accelerated gossip averaging procedure with a parameter R controlling the number of internal communication rounds and a contraction factor ρ that depends explicitly on R and the spectral gap. Our contribution is to embed this primitive into a stochastic accelerated optimization algorithm and to carry out a Lyapunov analysis that tracks both the disagreement recursion and the optimization recursion.

Accelerated decentralized methods and communication complexity. Beyond multi-consensus, several works have investigated acceleration in decentralized optimization, either via Nesterov-type momentum, dual acceleration, or primal–dual mirror maps [3, 4, 8, 12]. These methods often achieve communication complexity bounds that are optimal (or nearly optimal) in the deterministic strongly convex setting, with a characteristic $\sqrt{L/\mu}/\sqrt{1-\beta}$ dependence on the condition number and spectral gap. Extensions to time-varying or directed graphs further refine these guarantees under more general communication models [1, 6]. However, many of these results either assume exact oracles, lack fully explicit constants in the stochastic regime, or leave the relationship between the number of communication rounds per iteration and the stochastic batch size implicit.

Positioning of this work. The algorithm proposed in this paper, *Multi-Gossip Accelerated DSGD* (MC-ADSGD), combines several ingredients that have each appeared, in different forms, in the decentralized optimization literature: local stochastic gradients, multi-round accelerated gossip, and an accelerated extrapolation structure. Our emphasis here is not on claiming a universal improvement over all existing decentralized

methods, but on introducing this particular algorithmic design in the strongly convex stochastic regime and making the resulting communication-complexity dependence explicit. In particular, the analysis highlights how the gossip depth R , the spectral gap $1 - \beta$, and the stochastic noise level jointly enter the final bound, while the heterogeneity term contributes an explicit residual component.

3 Problem Setup

We consider the decentralized stochastic optimization problem

$$\min_{x \in \mathbb{R}^d} f(x) := \frac{1}{n} \sum_{i=1}^n f_i(x),$$

where node i has access only to its local objective f_i , its local stochastic gradient oracle, and messages received from its graph neighbors. For a stacked matrix variable $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$, we define the network average and disagreement operators by

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i, \quad \Pi := I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top.$$

The quantity ΠX measures consensus error. Throughout the paper, x^* denotes the unique minimizer of f , $f^* := f(x^*)$, and for each local objective we define $f_i^* := \min_x f_i(x)$. The heterogeneity measure is

$$G^* := f^* - \frac{1}{n} \sum_{i=1}^n f_i^*.$$

The quantity G^* captures heterogeneity at the optimum; in particular, $G^* = 0$ in the homogeneous-optimum case.

Our analysis in Section 5 is based on the following assumptions.

Assumption 1 (L -smoothness). Each local objective f_i is differentiable and L -smooth.

Assumption 2 (μ -strong convexity). Each local objective f_i is μ -strongly convex.

Assumption 3 (Unbiased stochastic gradients with bounded variance). The stochastic gradient oracle at node i is unbiased and has bounded conditional variance:

$$\mathbb{E}[\nabla F_i(x; \xi_i) | x] = \nabla f_i(x), \quad \mathbb{E}[\|\nabla F_i(x; \xi_i) - \nabla f_i(x)\|^2 | x] \leq \sigma^2.$$

Assumption 4 (Communication matrix). The communication matrix W is symmetric and doubly stochastic. Its spectral gap parameter is denoted by $\beta \in (0, 1]$.

Because the batch size and the communication depth are tied together in our final design, we set

$$B = R,$$

Algorithm 1 FastGossipAverage (FGA)**Require:** Local vector v_i , mixing matrix W , rounds R , acceleration parameter η **Ensure:** $\text{FGA}_R(v)_i$

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1: for all nodes  $i=1,\dots,n$  in parallel do
2:    $a_i^{(0)} \leftarrow v_i$ 
3:    $a_i^{(1)} \leftarrow \sum_{j=1}^n W_{ij}a_j^{(0)}$ 
4:   for  $r=1,2,\dots,R-1$  do
5:      $a_i^{(r+1)} \leftarrow (1+\eta)\sum_{j=1}^n W_{ij}a_j^{(r)} - \eta a_i^{(r-1)}$ 
6:   end for
7:   return  $a_i^{(R)}$ 
8: end for

```

where R is the number of accelerated gossip rounds used inside each outer iteration. The optimization parameter is

$$\theta = \sqrt{\frac{\mu\gamma}{2}}.$$

For the accelerated gossip operator, we use the effective contraction parameter

$$\rho := \sqrt{2}(1 - \sqrt{1-\beta})^R.$$

As R increases, the communication error decreases geometrically through ρ .

4 MC-ADSGD Algorithm

This section gives a concise specification of MC-ADSGD. We first state the accelerated gossip primitive used to suppress network disagreement, and then present the full iterative scheme together with its per-iteration oracle and communication costs.

Communication primitive: FastGossipAverage (FGA).

Main procedure: Multi-Gossip Accelerated DSGD. The iteration order is

stochastic gradient $\rightarrow$ z-update (FGA) $\rightarrow$ x-update (FGA) $\rightarrow$ y-extrapolation.

Design rationale and per-iteration complexity. The three state variables $(x^{(k)}, z^{(k)}, y^{(k)})$ have distinct algorithmic roles. The point $y^{(k)}$ is the query location for stochastic gradients, $z^{(k)}$ aggregates descent information and is consensus-refined, and $x^{(k)}$ tracks the primal progress. Accordingly, each outer iteration uses two FGA calls: one in the z-update and one in the x-update, followed by

$$y^{(k+1)} = (1-\theta)x^{(k+1)} + \theta z^{(k+1)}.$$

Algorithm 2 Multi-Gossip Accelerated DSGD (MC-ADSGD)**Require:** Stepsize γ , momentum parameter θ , gossip depth R

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1: Initialize  $x_i^{(0)} = y_i^{(0)} = z_i^{(0)} = 0$  for all  $i = 1, \dots, n$ 
2: for  $k = 0, 1, 2, \dots$  do
3:   for all nodes  $i = 1, \dots, n$  in parallel do
4:     Sample  $\xi_{i,k+1,1}, \dots, \xi_{i,k+1,R}$ 
5:      $g_i^{(k+1)} \leftarrow \frac{1}{R} \sum_{s=1}^R \nabla F_i(y_i^{(k)}; \xi_{i,k+1,s})$ 
6:      $z_i^{(k+1)} \leftarrow \text{FGA}_R\left(\frac{\frac{\theta}{\gamma} z^{(k)} + \mu y^{(k)} - g^{(k+1)}}{\frac{\theta}{\gamma} + \mu}\right)_i$ 
7:      $x_i^{(k+1)} \leftarrow \text{FGA}_R\left((1-\theta)x^{(k)} + \theta z^{(k+1)}\right)_i$ 
8:      $y_i^{(k+1)} \leftarrow (1-\theta)x_i^{(k+1)} + \theta z_i^{(k+1)}$ 
9:   end for
10: end for

```

At iteration k , node i forms

$$g_i^{(k+1)} = \frac{1}{R} \sum_{s=1}^R \nabla F_i(y_i^{(k)}; \xi_{i,k+1,s}),$$

which requires R stochastic-gradient oracle calls (hence nR calls over the network). The communication cost is exactly two depth- R FGA invocations, i.e., $2R$ gossip rounds per outer iteration. Therefore, after K outer iterations, the total communication budget is

$$T = 2RK,$$

which is the complexity variable used in Section 5.

5 Upper-Bound Analysis

This section presents the principal upper-bound theorem together with the supporting inequalities used in its proof. Detailed derivations are deferred to Appendix A. In addition to the heterogeneity quantity

$$G^* = f^* - \frac{1}{n} \sum_{i=1}^n f_i^*,$$

used throughout the paper, the derivation also employs the auxiliary quantity

$$\tilde{G} := \frac{1}{nL} \sum_{i=1}^n \|\nabla f_i(x^*)\|^2.$$

5.1 Primary upper-bound theorem

This subsection states the final communication-complexity guarantee in a compact form. The theorem below summarizes the dependence on optimization conditioning (L, μ) , network connectivity $(1 - \beta)$, stochastic noise level σ^2 , and the target accuracy ϵ . The explicit parameter construction used to realize this bound is deferred to Appendix A, so that the main text focuses on the final complexity consequence.

Theorem 5.1 (Convergence complexity for the strongly convex case). *Given $L \geq 0$, $n \geq 1$, $\beta \in [0, 1)$, and $\sigma > 0$, assume that the local objectives $\{f_i\}_{i=1}^n$ satisfy Assumptions 1 and 2 with $\mu > 0$, that the stochastic gradient oracles satisfy Assumption 3, and that the gossip matrix satisfies Assumption 4. Let*

$$\Delta_x := \|x^{(0)} - x^*\|^2.$$

Fix a target accuracy $\epsilon \in (0, L\Delta_x)$. Run Multi-Gossip Accelerated DSGD in Algorithm 2 with parameter choices (K, R, γ) specified in Appendix Theorem A.2. Then the total number of gossip rounds

$$T_\star(\epsilon)$$

needed to attain

$$\mathbb{E}[f(\bar{x}^{(K)}) - f(x^*)] \leq \epsilon$$

satisfies

$$T_\star(\epsilon) = \tilde{O}\left(\frac{\sigma^2}{\mu n \epsilon} \ln\left(\frac{L\Delta_x}{\epsilon}\right) + \sqrt{\frac{L/\mu}{1-\beta}} \ln\left(\frac{L\Delta_x}{\epsilon}\right)\right). \quad (5.1)$$

Here the $\tilde{O}(\cdot)$ notation suppresses logarithmic factors together with the heterogeneity-dependent contribution expressed in the main text through G^ .*

The proof of Theorem 5.1 proceeds through an explicit admissible-rate bound established in Appendix A. The auxiliary results below record the main ingredients that enter that proof.

5.2 Supporting lemmas

We next collect the structural estimates that drive the proof of Theorem 5.1. The roles of these lemmas are as follows: (i) relate the proof-internal heterogeneity quantity $\tilde{G}$ to the paper quantity G^* , (ii) control local model errors around decentralized surrogates, (iii) quantify stochastic variance reduction under batch size R , and (iv) establish coupled recursions for disagreement and optimization Lyapunov terms. Their full derivations are provided in Appendix A.

Lemma 5.1 (Relation between $\tilde{G}$ and G^*). *Under Assumption 1,*

$$\tilde{G} \leq 2G^*.$$

In particular, if $G^ = 0$, then $\tilde{G} = 0$.*

Lemma 5.2 (Local model bounds around decentralized surrogates). *Under Assumptions 1–2, define*

$$f(\bar{y}; y) := \frac{1}{n} \sum_{i=1}^n \left(f_i(y_i) + \langle \nabla f_i(y_i), \bar{y} - y_i \rangle \right),$$

and

$$D_f(x; y) := f(x) - f(\bar{y}; y) - \frac{1}{n} \sum_{i=1}^n \langle \nabla f_i(y_i), x - \bar{y} \rangle.$$

Then, for every $x \in \mathbb{R}^d$,

$$\begin{aligned} f(x) &\geq f(\bar{y}; y) + \frac{1}{n} \sum_{i=1}^n \langle \nabla f_i(y_i), x - \bar{y} \rangle + \frac{\mu}{2} \|x - \bar{y}\|^2 + \frac{\mu}{2n} \|\Pi y\|_F^2, \\ f(x) &\leq f(\bar{y}; y) + \frac{1}{n} \sum_{i=1}^n \langle \nabla f_i(y_i), x - \bar{y} \rangle + \frac{L}{2} \|x - \bar{y}\|^2 + \frac{L}{2n} \|\Pi y\|_F^2. \end{aligned}$$

Consequently,

$$\frac{\mu}{2} \|x - \bar{y}\|^2 + \frac{\mu}{2n} \|\Pi y\|_F^2 \leq D_f(x; y) \leq \frac{L}{2} \|x - \bar{y}\|^2 + \frac{L}{2n} \|\Pi y\|_F^2.$$

Lemma 5.3 (Conditional variance bounds). *Under Assumption 3, let*

$$h_i^{(k)} := \frac{1}{R} \sum_{s=1}^R \nabla F_i(y_i^{(k)}; \xi_{i,k,s}), \quad \bar{h}^{(k)} := \frac{1}{n} \sum_{i=1}^n h_i^{(k)},$$

and define

$$g^{(k)} := \mathbb{E}_k[\bar{h}^{(k)}] = \frac{1}{n} \sum_{i=1}^n \nabla f_i(y_i^{(k)}), \quad m^{(k+1)} := \mathbb{E}_k[\bar{z}^{(k+1)}].$$

Then

$$\mathbb{E}_k \|\bar{h}^{(k)} - g^{(k)}\|^2 \leq \frac{\sigma^2}{nR}, \quad \mathbb{E}_k \|\bar{z}^{(k+1)} - m^{(k+1)}\|^2 \leq \frac{\gamma^2 \sigma^2}{(\theta + \mu\gamma)^2 nR}.$$

Lemma 5.4 (Disagreement potential recursion). *Under Assumptions 1–4, define*

$$\Psi_k := \mathbb{E} [\|\Pi x^{(k)}\|_F^2] + \frac{4\rho^2(1+2\rho^2)}{(1-\rho^2)^2} \theta^2 \mathbb{E} [\|\Pi z^{(k)}\|_F^2],$$

and

$$\lambda := \max \left\{ 2(1-\theta)^2, \frac{(1-\rho^2)^2}{2\rho^2(1+2\rho^2)} \right\}.$$

Then, for all $k \geq 0$,

$$\begin{aligned} \Psi_{k+1} &\leq \left(\frac{3\rho^2}{1+2\rho^2} + \frac{144L^2\lambda\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \right) \Psi_k + \frac{72nL^2\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 \\ &\quad + \frac{144nL\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \mathbb{E} [f(\bar{x}^{(k)}) - f^*] + \frac{144nL\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \tilde{G} + \frac{24\rho^4(1+\rho^2)\gamma^2(n-1)\sigma^2}{(1-\rho^2)^3 R}. \end{aligned} \tag{5.2}$$

Lemma 5.5 (Optimization Lyapunov recursion). *Under Assumptions 1–4, define*

$$F_k := \mathbb{E}[f(\bar{x}^{(k)})] - f(x^*) + \left(\frac{\theta^2}{2\gamma} + \frac{\mu\theta}{2} \right) \mathbb{E} \left[\|\mathbb{E}_{k-1}[\bar{z}^{(k)}] - x^*\|^2 \right].$$

Then, for all $k \geq 0$,

$$\begin{aligned} F_{k+1} &\leq (1-\theta)F_k + \left(\frac{L}{2n} - (1-\theta)\frac{\mu}{2n} \right) \mathbb{E} \|\Pi y^{(k)}\|_F^2 - (1-\theta)\frac{\mu}{2} \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 \\ &\quad + \frac{\theta^2 \gamma \sigma^2}{2nR} \left(\frac{\gamma L + 1}{(\theta + \mu\gamma)^2} \right). \end{aligned} \quad (5.3)$$

Moreover, with

$$a_\Psi := \frac{\lambda}{n} (L - (1-\theta)\mu), \quad b_\Delta := \frac{(1-\theta)\mu}{2}, \quad \Delta_k := \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2,$$

and

$$D := \frac{\theta^2 \gamma \sigma^2}{2nR} \left(\frac{\gamma L + 1}{(\theta + \mu\gamma)^2} \right),$$

one has

$$F_{k+1} \leq (1-\theta)F_k + a_\Psi \Psi_k - b_\Delta \Delta_k + D.$$

6 Conclusion

We presented *Multi-Gossip Accelerated DSGD*, a decentralized stochastic optimization method that combines accelerated optimization dynamics with accelerated multi-round gossip communication. The main theorem gives an explicit strongly-convex convergence guarantee with transparent dependence on the stepsize, gossip depth, spectral gap, and stochastic variance. Under the homogeneous-optimum condition $G^* = 0$, the resulting communication complexity separates into a stochastic term of order $\sigma^2/(\mu n \epsilon)$ and a network term of order $\sqrt{L/\mu}/\sqrt{1-\beta}$, up to logarithmic factors.

From a methodological viewpoint, the analysis shows that optimization acceleration and communication acceleration should be designed together: the benefit of the extrapolation dynamics appears only when the disagreement recursion is controlled strongly enough by multi-gossip communication. This is the role of the FGA primitive and the explicit condition on

$$\rho = \sqrt{2}(1 - \sqrt{1-\beta})^R.$$

A natural next step is to extend the current analysis beyond the strongly convex regime and to establish analogous explicit rates under weaker convexity assumptions. Another important direction is to understand whether a similarly explicit accelerated analysis can be carried out under more general network models, including time-varying communication patterns.

Appendix

A Detailed Proofs for the Theoretical Results

This appendix presents the proof details in the order of the supplied theory file, while keeping the paper notation consistent with the main text.

Notation alignment. In the main text, the heterogeneity measure is

$$G^* := f^* - \frac{1}{n} \sum_{i=1}^n f_i^*, \quad f_i^* := \min_x f_i(x), \quad f^* := f(x^*).$$

The supplied proof file also uses the quantity

$$\tilde{G} := \frac{1}{nL} \sum_{i=1}^n \|\nabla f_i(x^*)\|^2,$$

which measures the mismatch of local gradients at the global minimizer. To avoid overloading the symbol G^* , we denote that proof-internal quantity by $\tilde{G}$ throughout this appendix.

Lemma A.1 (Relation between $\tilde{G}$ and the main-text heterogeneity G^*). *Assume each f_i is convex and L -smooth. Then*

$$\tilde{G} \leq 2G^*.$$

In particular, if $G^ = 0$, then $\tilde{G} = 0$.*

Proof. Let $x_i^* \in \arg\min_x f_i(x)$, so that $f_i(x_i^*) = f_i^*$ and $\nabla f_i(x_i^*) = 0$. By co-coercivity for L -smooth convex functions,

$$\|\nabla f_i(x^*) - \nabla f_i(x_i^*)\|^2 \leq 2L(f_i(x^*) - f_i(x_i^*) - \langle \nabla f_i(x_i^*), x^* - x_i^* \rangle) = 2L(f_i(x^*) - f_i^*).$$

Summing over i and dividing by nL yields

$$\tilde{G} = \frac{1}{nL} \sum_{i=1}^n \|\nabla f_i(x^*)\|^2 \leq \frac{2}{n} \sum_{i=1}^n (f_i(x^*) - f_i^*) = 2 \left(f^* - \frac{1}{n} \sum_{i=1}^n f_i^* \right) = 2G^*.$$

If $G^* = 0$, then the right-hand side is zero, hence $\tilde{G} = 0$. □

A.1 Setup and assumptions

We consider

$$\min_{x \in \mathbb{R}^d} f(x) := \frac{1}{n} \sum_{i=1}^n f_i(x).$$

For a stacked matrix $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$, define

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i, \quad \Pi := I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top,$$

so that ΠX measures disagreement. We write

$$\bar{x}^{(k)} := \frac{1}{n} \sum_{i=1}^n x_i^{(k)}, \quad \bar{y}^{(k)} := \frac{1}{n} \sum_{i=1}^n y_i^{(k)}, \quad \bar{z}^{(k)} := \frac{1}{n} \sum_{i=1}^n z_i^{(k)}.$$

The algorithm parameters are

$$\theta := \sqrt{\frac{\mu\gamma}{2}}, \quad \rho := \sqrt{2}(1 - \sqrt{1 - \beta})^R, \quad B = R.$$

Assumption 5 (L -smoothness). Each local objective f_i is differentiable and L -smooth.

Assumption 6 (μ -strong convexity). The global objective f is μ -strongly convex.

Assumption 7 (Unbiased stochastic gradients with bounded variance). For each node i ,

$$\mathbb{E}[\nabla F_i(x; \xi_i) | x] = \nabla f_i(x), \quad \mathbb{E}[\|\nabla F_i(x; \xi_i) - \nabla f_i(x)\|^2 | x] \leq \sigma^2.$$

Assumption 8 (Communication matrix and accelerated gossip). The mixing matrix W is symmetric and doubly stochastic. The accelerated gossip operator FGA_R satisfies

$$\|\Pi \text{FGA}_R(U)\|_F^2 \leq \rho^2 \|\Pi U\|_F^2, \quad \rho = \sqrt{2}(1 - \sqrt{1 - \beta})^R,$$

for every stacked matrix U .

Because FGA_R is a polynomial in a doubly stochastic matrix, it also preserves averages. Hence the averaged iterates satisfy

$$\bar{z}^{(k+1)} = \frac{\frac{\theta}{\gamma} \bar{z}^{(k)} + \mu \bar{y}^{(k)} - \bar{g}^{(k+1)}}{\frac{\theta}{\gamma} + \mu}, \quad \bar{x}^{(k+1)} = (1 - \theta) \bar{x}^{(k)} + \theta \bar{z}^{(k+1)}, \quad \bar{y}^{(k)} = (1 - \theta) \bar{x}^{(k)} + \theta \bar{z}^{(k)}.$$

A.2 Recursion for the disagreement potential

Lemma A.2 (Recursion of the disagreement potential). *Define*

$$\Psi_k := \mathbb{E}[\|\Pi x^{(k)}\|_F^2] + \frac{4\rho^2(1 + 2\rho^2)}{(1 - \rho^2)^2} \theta^2 \mathbb{E}[\|\Pi z^{(k)}\|_F^2],$$

and

$$\lambda := \max \left\{ 2(1 - \theta)^2, \frac{(1 - \rho^2)^2}{2\rho^2(1 + 2\rho^2)} \right\}.$$

Then, for all $k \geq 0$,

$$\begin{aligned} \Psi_{k+1} \leq & \left(\frac{3\rho^2}{1+2\rho^2} + \frac{144L^2\lambda\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \right) \Psi_k + \frac{72nL^2\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 \\ & + \frac{144nL\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \mathbb{E} [f(\bar{x}^{(k)}) - f^*] + \frac{144nL\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \tilde{G} + \frac{24\rho^4(1+\rho^2)\gamma^2(n-1)\sigma^2}{(1-\rho^2)^3 R}. \end{aligned} \quad (\text{A.1})$$

Proof. Set

$$A := \frac{\theta}{\gamma} + \mu, \quad U^{(k+1)} := \frac{\frac{\theta}{\gamma} z^{(k)} + \mu y^{(k)} - g^{(k+1)}}{A}, \quad V^{(k+1)} := (1-\theta)x^{(k)} + \theta z^{(k+1)}.$$

Then

$$z^{(k+1)} = \text{FGA}_R(U^{(k+1)}), \quad x^{(k+1)} = \text{FGA}_R(V^{(k+1)}), \quad y^{(k)} = (1-\theta)x^{(k)} + \theta z^{(k)}.$$

Applying the accelerated-gossip contraction to the z -update gives

$$\|\Pi z^{(k+1)}\|_F^2 \leq \frac{\rho^2}{A^2} \left\| \frac{\theta}{\gamma} \Pi z^{(k)} + \mu \Pi y^{(k)} - \Pi g^{(k+1)} \right\|_F^2.$$

Using $\|a+b+c\|^2 \leq 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$ and conditional expectation,

$$\mathbb{E}_k \|\Pi z^{(k+1)}\|_F^2 \leq \frac{3\rho^2}{A^2} \left(\frac{\theta^2}{\gamma^2} \|\Pi z^{(k)}\|_F^2 + \mu^2 \|\Pi y^{(k)}\|_F^2 + \mathbb{E}_k \|\Pi g^{(k+1)}\|_F^2 \right). \quad (\text{A.2})$$

Write

$$g^{(k+1)} = \nabla f(y^{(k)}) + \zeta^{(k+1)}, \quad \mathbb{E}_k[\zeta^{(k+1)}] = 0,$$

where $\nabla f(y^{(k)})$ denotes the stacked matrix $[\nabla f_1(y_1^{(k)}), \dots, \nabla f_n(y_n^{(k)})]$. Then

$$\mathbb{E}_k \|\Pi g^{(k+1)}\|_F^2 = \|\Pi \nabla f(y^{(k)})\|_F^2 + \mathbb{E}_k \|\Pi \zeta^{(k+1)}\|_F^2.$$

For the deterministic term,

$$\|\Pi \nabla f(y^{(k)})\|_F^2 \leq 2 \sum_{i=1}^n \|\nabla f_i(y_i^{(k)}) - \nabla f_i(\bar{y}^{(k)})\|^2 + 2 \|\Pi \nabla f(\bar{y}^{(k)})\|_F^2.$$

By L -smoothness,

$$\sum_{i=1}^n \|\nabla f_i(y_i^{(k)}) - \nabla f_i(\bar{y}^{(k)})\|^2 \leq L^2 \|\Pi y^{(k)}\|_F^2.$$

Also,

$$\|\Pi \nabla f(\bar{y}^{(k)})\|_F^2 \leq 2 \|\Pi(\nabla f(\bar{y}^{(k)}) - \nabla f(\bar{x}^{(k)}))\|_F^2 + 2 \|\Pi \nabla f(\bar{x}^{(k)})\|_F^2,$$

with

$$\|\Pi(\nabla f(\bar{y}^{(k)}) - \nabla f(\bar{x}^{(k)}))\|_F^2 \leq nL^2 \|\bar{y}^{(k)} - \bar{x}^{(k)}\|^2.$$

For the remaining term, use $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ with $a = \nabla f_i(\bar{x}^{(k)}) - \nabla f_i(x^*)$ and $b = \nabla f_i(x^*)$, and then the co-coercivity inequality for convex L -smooth functions:

$$\|\nabla f_i(\bar{x}^{(k)}) - \nabla f_i(x^*)\|^2 \leq 2L \left(f_i(\bar{x}^{(k)}) - f_i(x^*) - \langle \nabla f_i(x^*), \bar{x}^{(k)} - x^* \rangle \right).$$

Summing over i and using $\sum_{i=1}^n \nabla f_i(x^*) = 0$ gives

$$\sum_{i=1}^n \|\nabla f_i(\bar{x}^{(k)})\|^2 \leq 4nL(f(\bar{x}^{(k)}) - f^*) + 2nL\tilde{G}.$$

Therefore,

$$\|\Pi \nabla f(y^{(k)})\|_F^2 \leq 2L^2 \|\Pi y^{(k)}\|_F^2 + 4nL^2 \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 + 16nL(f(\bar{x}^{(k)}) - f^*) + 8nL\tilde{G}. \quad (\text{A.3})$$

For the stochastic part, since each node uses a mini-batch of size R ,

$$\mathbb{E}_k \|\zeta_i^{(k+1)}\|^2 \leq \frac{\sigma^2}{R}.$$

Let $\bar{\zeta}^{(k+1)} := \frac{1}{n} \sum_{i=1}^n \zeta_i^{(k+1)}$. Then

$$\|\Pi \bar{\zeta}^{(k+1)}\|_F^2 = \sum_{i=1}^n \|\zeta_i^{(k+1)}\|^2 - n \|\bar{\zeta}^{(k+1)}\|^2.$$

Using conditional independence and zero conditional mean,

$$\mathbb{E}_k \|\bar{\zeta}^{(k+1)}\|^2 = \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}_k \|\zeta_i^{(k+1)}\|^2,$$

and hence

$$\mathbb{E}_k \|\Pi \bar{\zeta}^{(k+1)}\|_F^2 \leq \frac{(n-1)\sigma^2}{R}. \quad (\text{A.4})$$

Substituting (A.3) and (A.4) into (A.2) gives

$$\begin{aligned} \mathbb{E}_k \|\Pi z^{(k+1)}\|_F^2 &\leq \frac{3\rho^2(\mu^2 + 2L^2)}{A^2} \|\Pi y^{(k)}\|_F^2 + \frac{12nL^2\rho^2}{A^2} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 \\ &\quad + \frac{48nL\rho^2}{A^2} (f(\bar{x}^{(k)}) - f^*) + \frac{24nL\rho^2}{A^2} \tilde{G} + \frac{3\rho^2(n-1)\sigma^2}{A^2R} + \frac{3\rho^2\theta^2}{A^2\gamma^2} \|\Pi z^{(k)}\|_F^2. \end{aligned} \quad (\text{A.5})$$

Next, apply the accelerated-gossip contraction to the x -update:

$$\|\Pi x^{(k+1)}\|_F^2 \leq \rho^2 \|(1-\theta)\Pi x^{(k)} + \theta \Pi z^{(k+1)}\|_F^2 \leq 2\rho^2(1-\theta)^2 \|\Pi x^{(k)}\|_F^2 + 2\rho^2\theta^2 \|\Pi z^{(k+1)}\|_F^2.$$

Also,

$$\Pi y^{(k)} = (1-\theta)\Pi x^{(k)} + \theta\Pi z^{(k)},$$

so

$$\|\Pi y^{(k)}\|_F^2 \leq 2(1-\theta)^2 \|\Pi x^{(k)}\|_F^2 + 2\theta^2 \|\Pi z^{(k)}\|_F^2 \leq \lambda \Psi_k.$$

Now multiply (A.5) by

$$\alpha := \frac{4\rho^2(1+2\rho^2)}{(1-\rho^2)^2} \theta^2$$

and add the result to the bound for $\|\Pi x^{(k+1)}\|_F^2$. Before simplification this gives the same intermediate recursion as in the supplied proof file. Finally use

$$A \geq \frac{\theta}{\gamma}, \quad A^{-2} \leq \frac{\gamma^2}{\theta^2}, \quad 1+2\rho^2 \leq 2(1+\rho^2), \quad \frac{1}{(1-\rho^2)^2} \leq \frac{2}{(1-\rho^2)^3},$$

and absorb the $\|\Pi y^{(k)}\|_F^2$ term with the bound $\|\Pi y^{(k)}\|_F^2 \leq \lambda \Psi_k$. This yields exactly (A.1). $\square$

A.3 Local surrogate bounds and the optimization Lyapunov recursion

Lemma A.3 (Local model bounds around decentralized surrogates). *Define*

$$f(\bar{y}; y) := \frac{1}{n} \sum_{i=1}^n \left(f_i(y_i) + \langle \nabla f_i(y_i), \bar{y} - y_i \rangle \right),$$

and

$$D_f(x; y) := f(x) - f(\bar{y}; y) - \frac{1}{n} \sum_{i=1}^n \langle \nabla f_i(y_i), x - \bar{y} \rangle.$$

Then, for every $x \in \mathbb{R}^d$,

$$\begin{aligned} f(x) &\geq f(\bar{y}; y) + \frac{1}{n} \sum_{i=1}^n \langle \nabla f_i(y_i), x - \bar{y} \rangle + \frac{\mu}{2} \|x - \bar{y}\|^2 + \frac{\mu}{2n} \|\Pi y\|_F^2, \\ f(x) &\leq f(\bar{y}; y) + \frac{1}{n} \sum_{i=1}^n \langle \nabla f_i(y_i), x - \bar{y} \rangle + \frac{L}{2} \|x - \bar{y}\|^2 + \frac{L}{2n} \|\Pi y\|_F^2. \end{aligned}$$

Consequently,

$$\frac{\mu}{2} \|x - \bar{y}\|^2 + \frac{\mu}{2n} \|\Pi y\|_F^2 \leq D_f(x; y) \leq \frac{L}{2} \|x - \bar{y}\|^2 + \frac{L}{2n} \|\Pi y\|_F^2.$$

Proof. Apply the μ -strong convexity and L -smoothness inequalities to each f_i at the pair (x, y_i) , average over i , and use the identity

$$\frac{1}{n} \sum_{i=1}^n \|x - y_i\|^2 = \|x - \bar{y}\|^2 + \frac{1}{n} \|\Pi y\|_F^2.$$

The claimed two-sided bound for $D_f(x; y)$ follows immediately. $\square$

Lemma A.4 (Conditional variance bounds for averaged gradients and averaged z-updates).

Let

$$h_i^{(k)} := \frac{1}{R} \sum_{s=1}^R \nabla F_i(y_i^{(k)}; \zeta_{i,k,s}), \quad \bar{h}^{(k)} := \frac{1}{n} \sum_{i=1}^n h_i^{(k)},$$

and define

$$g^{(k)} := \mathbb{E}_k[\bar{h}^{(k)}] = \frac{1}{n} \sum_{i=1}^n \nabla f_i(y_i^{(k)}), \quad m^{(k+1)} := \mathbb{E}_k[\bar{z}^{(k+1)}].$$

Then

$$\mathbb{E}_k \|\bar{h}^{(k)} - g^{(k)}\|^2 \leq \frac{\sigma^2}{nR}, \quad \mathbb{E}_k \|\bar{z}^{(k+1)} - m^{(k+1)}\|^2 \leq \frac{\gamma^2 \sigma^2}{(\theta + \mu\gamma)^2 nR}.$$

Proof. The first estimate follows from mini-batch variance reduction and conditional independence across nodes. For the second estimate, use the averaged z-update

$$\bar{z}^{(k+1)} = \frac{\theta \bar{z}^{(k)} + \mu\gamma \bar{y}^{(k)} - \gamma \bar{h}^{(k)}}{\theta + \mu\gamma},$$

so that

$$\bar{z}^{(k+1)} - m^{(k+1)} = -\frac{\gamma}{\theta + \mu\gamma} (\bar{h}^{(k)} - g^{(k)}).$$

Squaring and taking conditional expectation gives the result. $\square$

Lemma A.5 (Recursion of the optimization Lyapunov term). Define

$$F_k := \mathbb{E}[f(\bar{x}^{(k)})] - f(x^*) + \left(\frac{\theta^2}{2\gamma} + \frac{\mu\theta}{2} \right) \mathbb{E} \left[\|\mathbb{E}_{k-1}[\bar{z}^{(k)}] - x^*\|^2 \right].$$

Then, for all $k \geq 0$,

$$\begin{aligned} F_{k+1} &\leq (1-\theta)F_k + \left(\frac{L}{2n} - (1-\theta)\frac{\mu}{2n} \right) \mathbb{E} \|\Pi y^{(k)}\|_F^2 - (1-\theta)\frac{\mu}{2} \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 \\ &\quad + \frac{\theta^2 \gamma \sigma^2}{2nR} \left(\frac{\gamma L + 1}{(\theta + \mu\gamma)^2} \right). \end{aligned} \tag{A.6}$$

Moreover, using $\|\Pi y^{(k)}\|_F^2 \leq \lambda \Psi_k$, one has

$$F_{k+1} \leq (1-\theta)F_k + a_\Psi \Psi_k - b_\Delta \Delta_k + D, \tag{A.7}$$

where

$$a_\Psi := \frac{\lambda}{n} (L - (1-\theta)\mu), \quad b_\Delta := \frac{(1-\theta)\mu}{2}, \quad \Delta_k := \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2,$$

and

$$D := \frac{\theta^2 \gamma \sigma^2}{2nR} \left(\frac{\gamma L + 1}{(\theta + \mu\gamma)^2} \right).$$

Proof. By average preservation of accelerated gossip,

$$\bar{y}^{(k)} = (1-\theta)\bar{x}^{(k)} + \theta\bar{z}^{(k)}, \quad \bar{x}^{(k+1)} = (1-\theta)\bar{x}^{(k)} + \theta\bar{z}^{(k+1)},$$

so

$$\bar{x}^{(k+1)} - \bar{y}^{(k)} = \theta(\bar{z}^{(k+1)} - \bar{z}^{(k)}).$$

Applying the upper bound from Lemma A.3 at $x = \bar{x}^{(k+1)}$ and $y = \bar{y}^{(k)}$ gives

$$\begin{aligned} f(\bar{x}^{(k+1)}) &\leq f(\bar{y}^{(k)}; \bar{y}^{(k)}) + \theta \left\langle \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{y}_i^{(k)}), \bar{z}^{(k+1)} - \bar{z}^{(k)} \right\rangle \\ &\quad + \frac{L\theta^2}{2} \|\bar{z}^{(k+1)} - \bar{z}^{(k)}\|^2 + \frac{L}{2n} \|\Pi \bar{y}^{(k)}\|_F^2. \end{aligned} \quad (\text{A.8})$$

Insert x^* by splitting the inner product:

$$\bar{z}^{(k+1)} - \bar{z}^{(k)} = (x^* - \bar{z}^{(k)}) + (\bar{z}^{(k+1)} - x^*).$$

Let

$$g^{(k)} := \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{y}_i^{(k)}), \quad m^{(k+1)} := \mathbb{E}_k[\bar{z}^{(k+1)}].$$

Taking conditional expectation in (A.8) and using Lemma A.4,

$$\begin{aligned} \mathbb{E}_k[f(\bar{x}^{(k+1)})] &\leq f(\bar{y}^{(k)}; \bar{y}^{(k)}) + \theta \langle g^{(k)}, x^* - \bar{z}^{(k)} \rangle + \theta \langle g^{(k)}, m^{(k+1)} - x^* \rangle \\ &\quad + \frac{L\theta^2}{2} \|m^{(k+1)} - \bar{z}^{(k)}\|^2 + \frac{L}{2n} \|\Pi \bar{y}^{(k)}\|_F^2 + \frac{L\theta^2 \gamma^2 \sigma^2}{2(\theta + \mu\gamma)^2 nR}. \end{aligned} \quad (\text{A.9})$$

Next, use

$$\theta(x^* - \bar{z}^{(k)}) = \theta(x^* - \bar{y}^{(k)}) + (1-\theta)(\bar{x}^{(k)} - \bar{y}^{(k)}).$$

Applying the lower bound of Lemma A.3 with $x = x^*$ and $x = \bar{x}^{(k)}$ gives

$$\begin{aligned} f(\bar{y}^{(k)}; \bar{y}^{(k)}) + \theta \langle g^{(k)}, x^* - \bar{z}^{(k)} \rangle &\leq \theta f(x^*) - \frac{\mu\theta}{2} \|\bar{y}^{(k)} - x^*\|^2 - \frac{\mu\theta}{2n} \|\Pi \bar{y}^{(k)}\|_F^2 \\ &\quad + (1-\theta)(f(\bar{x}^{(k)}) - D_f(\bar{x}^{(k)}; \bar{y}^{(k)})). \end{aligned}$$

Substituting this into (A.9) yields

$$\begin{aligned} \mathbb{E}_k[f(\bar{x}^{(k+1)})] &\leq (1-\theta)(f(\bar{x}^{(k)}) - D_f(\bar{x}^{(k)}; \bar{y}^{(k)})) + \theta f(x^*) - \frac{\mu\theta}{2} \|\bar{y}^{(k)} - x^*\|^2 \\ &\quad - \frac{\mu\theta}{2n} \|\Pi \bar{y}^{(k)}\|_F^2 + \theta \langle g^{(k)}, m^{(k+1)} - x^* \rangle + \frac{L\theta^2}{2} \|m^{(k+1)} - \bar{z}^{(k)}\|^2 \end{aligned} \quad (\text{A.10})$$

$$+ \frac{L}{2n} \|\Pi \bar{y}^{(k)}\|_F^2 + \frac{L\theta^2 \gamma^2 \sigma^2}{2(\theta + \mu\gamma)^2 nR}. \quad (\text{A.11})$$

The averaged z-update identity implies

$$g^{(k)} + \frac{\theta}{\gamma}(m^{(k+1)} - \bar{z}^{(k)}) + \mu(m^{(k+1)} - \bar{y}^{(k)}) = 0.$$

Hence

$$\begin{aligned} \theta \langle g^{(k)}, m^{(k+1)} - x^* \rangle &= -\frac{\theta^2}{\gamma} \langle m^{(k+1)} - \bar{z}^{(k)}, m^{(k+1)} - x^* \rangle \\ &\quad - \mu \theta \langle m^{(k+1)} - \bar{y}^{(k)}, m^{(k+1)} - x^* \rangle. \end{aligned}$$

Using

$$2\langle a-b, a-c \rangle = \|a-b\|^2 + \|a-c\|^2 - \|b-c\|^2,$$

we obtain

$$\begin{aligned} \theta \langle g^{(k)}, m^{(k+1)} - x^* \rangle &= \frac{\theta^2}{2\gamma} \|\bar{z}^{(k)} - x^*\|^2 - \frac{\theta^2}{2\gamma} \|m^{(k+1)} - x^*\|^2 - \frac{\theta^2}{2\gamma} \|m^{(k+1)} - \bar{z}^{(k)}\|^2 \\ &\quad + \frac{\mu\theta}{2} \|\bar{y}^{(k)} - x^*\|^2 - \frac{\mu\theta}{2} \|m^{(k+1)} - x^*\|^2 - \frac{\mu\theta}{2} \|m^{(k+1)} - \bar{y}^{(k)}\|^2. \end{aligned}$$

Substituting this into (A.11), canceling the $\|\bar{y}^{(k)} - x^*\|^2$ term, and discarding the nonpositive term $-\frac{\mu\theta}{2} \|m^{(k+1)} - \bar{y}^{(k)}\|^2$, we get

$$\begin{aligned} \mathbb{E}_k[f(\bar{x}^{(k+1)})] - f(x^*) &+ \left(\frac{\theta^2}{2\gamma} + \frac{\mu\theta}{2} \right) \|m^{(k+1)} - x^*\|^2 \\ &\leq (1-\theta)(f(\bar{x}^{(k)}) - f(x^*)) + \frac{\theta^2}{2\gamma} \|\bar{z}^{(k)} - x^*\|^2 - \left(\frac{\theta^2}{2\gamma} - \frac{L\theta^2}{2} \right) \|m^{(k+1)} - \bar{z}^{(k)}\|^2 \\ &\quad - (1-\theta)D_f(\bar{x}^{(k)}; y^{(k)}) + \left(\frac{L}{2n} - \frac{\mu\theta}{2n} \right) \|\Pi y^{(k)}\|_F^2 + \frac{L\theta^2\gamma^2\sigma^2}{2(\theta+\mu\gamma)^2nR}. \end{aligned}$$

Now use the lower bound from Lemma A.3:

$$D_f(\bar{x}^{(k)}; y^{(k)}) \geq \frac{\mu}{2} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 + \frac{\mu}{2n} \|\Pi y^{(k)}\|_F^2.$$

Then

$$\begin{aligned} &-(1-\theta)D_f(\bar{x}^{(k)}; y^{(k)}) + \frac{L}{2n} \|\Pi y^{(k)}\|_F^2 \\ &\leq -(1-\theta)\frac{\mu}{2} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2 + \left(\frac{L}{2n} - (1-\theta)\frac{\mu}{2n} \right) \|\Pi y^{(k)}\|_F^2. \end{aligned}$$

Finally, by Lemma A.4,

$$\mathbb{E} \|\bar{z}^{(k)} - x^*\|^2 \leq \mathbb{E} \|\mathbb{E}_{k-1}[\bar{z}^{(k)}] - x^*\|^2 + \frac{\gamma^2\sigma^2}{(\theta+\mu\gamma)^2nR}.$$

Taking full expectation and collecting the two variance contributions gives (A.6). Replacing $\|\Pi y^{(k)}\|_F^2$ by $\lambda \Psi_k$ yields (A.7). $\square$

A.4 Combined Lyapunov recursion and unrolling

Lemma A.6 (Combined Lyapunov recursion). *Let*

$$\Phi_k := F_k + u\Psi_k, \quad \Delta_k := \mathbb{E} \|\bar{x}^{(k)} - \bar{y}^{(k)}\|^2.$$

Set

$$\begin{aligned} a_\Psi &:= \frac{\lambda}{n} (L - (1-\theta)\mu), & b_\Delta &:= \frac{(1-\theta)\mu}{2}, \\ q_\Psi &:= \frac{3\rho^2}{1+2\rho^2} + \frac{144L^2\lambda\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3}, & c_\Delta &:= \frac{72nL^2\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3}, & c_F &:= \frac{144nL\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3}, \\ C &:= c_F\tilde{G} + \frac{24\rho^4(1+\rho^2)\gamma^2(n-1)\sigma^2}{(1-\rho^2)^3R}, & D &:= \frac{\theta^2\gamma\sigma^2}{2nR} \left(\frac{\gamma L + 1}{(\theta + \mu\gamma)^2} \right). \end{aligned}$$

Assume

$$\delta := 1 - \frac{\theta}{2} - q_\Psi > 0, \quad \frac{[a_\Psi]_+}{\delta} \leq u \leq \min \left\{ \frac{\theta}{2c_F}, \frac{b_\Delta}{c_\Delta} \right\}.$$

Then, for all $k \geq 0$,

$$\Phi_{k+1} \leq \left(1 - \frac{\theta}{2}\right) \Phi_k + uC + D.$$

Proof. From Lemma A.5,

$$F_{k+1} \leq (1-\theta)F_k + a_\Psi\Psi_k - b_\Delta\Delta_k + D.$$

From Lemma A.2 and the fact that $\mathbb{E}[f(\bar{x}^{(k)}) - f^*] \leq F_k$,

$$\Psi_{k+1} \leq q_\Psi\Psi_k + c_\Delta\Delta_k + c_FF_k + C.$$

Multiply the second inequality by u and add it to the first one. This yields

$$\Phi_{k+1} \leq (1-\theta+uc_F)F_k + (a_\Psi+uq_\Psi)\Psi_k + (uc_\Delta-b_\Delta)\Delta_k + uC + D.$$

Now the upper bound $u \leq \theta/(2c_F)$ implies

$$(1-\theta+uc_F)F_k \leq \left(1 - \frac{\theta}{2}\right) F_k.$$

Next, by the definition of δ and the lower bound $u \geq [a_\Psi]_+/\delta$,

$$a_\Psi + uq_\Psi \leq u \left(1 - \frac{\theta}{2}\right),$$

so

$$(a_\Psi + uq_\Psi)\Psi_k \leq \left(1 - \frac{\theta}{2}\right) u\Psi_k.$$

Finally, the upper bound $u \leq b_\Delta / c_\Delta$ gives $uc_\Delta - b_\Delta \leq 0$, so the Δ_k term is nonpositive and may be dropped. Therefore,

$$\Phi_{k+1} \leq \left(1 - \frac{\theta}{2}\right) (F_k + u\Psi_k) + uC + D = \left(1 - \frac{\theta}{2}\right) \Phi_k + uC + D.$$

□

Proposition A.1 (Feasibility of the interval for u). *Assume $\rho < 1$. Then there exists $\bar{\gamma}(R) > 0$ such that for every $0 < \gamma \leq \bar{\gamma}(R)$ one has $\delta > 0$ and the interval*

$$\frac{[a_\Psi]_+}{\delta} \leq u \leq \min \left\{ \frac{\theta}{2c_F}, \frac{b_\Delta}{c_\Delta} \right\}$$

is nonempty.

Proof. For fixed R with $\rho < 1$, the quantity

$$\lambda = \max \left\{ 2(1-\theta)^2, \frac{(1-\rho^2)^2}{2\rho^2(1+2\rho^2)} \right\}$$

remains bounded as $\gamma \downarrow 0$, because $\theta = \sqrt{\mu\gamma/2} \rightarrow 0$. Hence

$$q_\Psi = \frac{3\rho^2}{1+2\rho^2} + O(\gamma^2), \quad \delta = 1 - \frac{\theta}{2} - q_\Psi = \frac{1-\rho^2}{1+2\rho^2} + o(1).$$

Since $\rho < 1$, we have $(1-\rho^2)/(1+2\rho^2) > 0$, and therefore $\delta > 0$ for all sufficiently small $\gamma > 0$. Also,

$$a_\Psi = \frac{\lambda}{n} (L - (1-\theta)\mu) = O(1),$$

so the lower endpoint $[a_\Psi]_+ / \delta$ stays bounded. On the other hand,

$$\frac{\theta}{2c_F} = \frac{(1-\rho^2)^3}{288nL\rho^4(1+\rho^2)\gamma^2} \theta \asymp \gamma^{-3/2} \rightarrow +\infty,$$

while

$$\frac{b_\Delta}{c_\Delta} = \frac{(1-\theta)\mu(1-\rho^2)^3}{144nL^2\rho^4(1+\rho^2)\gamma^2} \asymp \gamma^{-2} \rightarrow +\infty.$$

Thus the right endpoint diverges to $+\infty$ whereas the left endpoint remains bounded, so the interval is nonempty for all sufficiently small γ . □

Corollary A.1 (Unrolled Lyapunov bound). *Assume the conditions of Lemma A.6, and let u satisfy the stated interval condition. Then, for every $k \geq 0$,*

$$\Phi_k \leq \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \frac{2}{\theta} \left(1 - \left(1 - \frac{\theta}{2}\right)^k\right) \mathcal{N}_{R,\gamma},$$

where

$$\mathcal{N}_{R,\gamma} := u \left(\frac{144nL\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \tilde{G} + \frac{24\rho^4(1+\rho^2)\gamma^2(n-1)\sigma^2}{(1-\rho^2)^3 R} \right) + \frac{\mu\gamma^2\sigma^2}{4nR} \left(\frac{\gamma L + 1}{(\sqrt{\mu\gamma/2} + \mu\gamma)^2} \right).$$

In particular,

$$\Phi_k \leq \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \frac{2}{\theta} \mathcal{N}_{R,\gamma}.$$

Proof. Iterating the one-step recursion from Lemma A.6 gives

$$\Phi_k \leq \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \sum_{t=0}^{k-1} \left(1 - \frac{\theta}{2}\right)^{k-1-t} (uC + D).$$

Since $uC + D$ does not depend on t , the geometric-series identity yields

$$\Phi_k \leq \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \frac{1 - \left(1 - \frac{\theta}{2}\right)^k}{1 - \left(1 - \frac{\theta}{2}\right)} (uC + D) = \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \frac{2}{\theta} \left(1 - \left(1 - \frac{\theta}{2}\right)^k\right) (uC + D).$$

Substituting the definitions of C and D shows that $uC + D = \mathcal{N}_{R,\gamma}$. $\square$

A.5 Explicit admissible parameters and the final rate

Theorem A.1 (Detailed convergence bound in the notation of the supplied proof). *Assume the conditions of Proposition A.1, and choose $u > 0$ satisfying the interval condition in Lemma A.6. Then, for every $k \geq 0$,*

$$F_k \leq \Phi_k \leq \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \frac{2}{\theta} \mathcal{N}_{R,\gamma},$$

where

$$\theta = \sqrt{\frac{\mu\gamma}{2}}, \quad \rho = \sqrt{2}(1 - \sqrt{1 - \beta})^R.$$

Consequently,

$$\mathbb{E}[f(\bar{x}^{(k)}) - f(x^*)] \leq \left(1 - \frac{\theta}{2}\right)^k \Phi_0 + \frac{2}{\theta} \mathcal{N}_{R,\gamma}.$$

Proof. Since $\Phi_k = F_k + u\Psi_k$ and $u\Psi_k \geq 0$, one has $F_k \leq \Phi_k$. The claim therefore follows directly from Corollary A.1 and the definition of F_k . $\square$

Theorem A.2 (Explicit admissible choice of (γ, R)). *Assume Assumptions 1–4. Let*

$$0 < \gamma \leq \frac{1}{8L}, \quad \theta = \sqrt{\frac{\mu\gamma}{2}}, \quad \rho^2 = 2(1 - \sqrt{1 - \beta})^{2R} \leq \frac{\mu\gamma}{4}.$$

Let

$$\Delta_x := \|x^{(0)} - x^*\|^2.$$

Then the interval for u in Lemma A.6 is nonempty. Moreover, with the choice

$$u := \frac{a_\Psi}{\delta},$$

for every $k \geq 0$ one has

$$\mathbb{E}[f(\bar{x}^{(k)}) - f(x^*)] \leq \left(1 - \frac{\theta}{2}\right)^k L\Delta_x + \frac{192}{119}\theta\tilde{G} + \frac{n-1}{6nLR}\sigma^2 + \frac{9\sqrt{2}}{8}\frac{\sigma^2}{nR}\sqrt{\frac{\gamma}{\mu}}. \quad (\text{A.12})$$

Proof. By the assumptions on γ and R ,

$$\rho^2 \leq \frac{\mu\gamma}{4} \leq \frac{\mu}{32L} \leq \frac{1}{32}.$$

Hence

$$\lambda = \frac{(1-\rho^2)^2}{2\rho^2(1+2\rho^2)},$$

because for $\rho^2 \leq 1/32$ the second term in the definition of λ dominates $2(1-\theta)^2$. Using

$$\frac{1+\rho^2}{(1+2\rho^2)(1-\rho^2)} \leq \frac{33}{31}, \quad \gamma \leq \frac{1}{8L},$$

we obtain

$$\begin{aligned} q_\Psi &= \frac{3\rho^2}{1+2\rho^2} + \frac{72L^2\gamma^2\rho^2(1+\rho^2)}{(1+2\rho^2)(1-\rho^2)} \\ &\leq 3\rho^2 + \frac{9}{8} \cdot \frac{33}{31}\rho^2 < 3\rho^2 + \frac{6}{5}\rho^2 = \frac{21}{5}\rho^2 \leq \frac{21}{20}\mu\gamma \leq \frac{21}{160}. \end{aligned}$$

Therefore,

$$\delta = 1 - \frac{\theta}{2} - q_\Psi \geq 1 - \frac{1}{8} - \frac{21}{160} = \frac{119}{160}.$$

Choose $u = a_\Psi/\delta$. Since $a_\Psi \geq 0$, this satisfies the lower bound. We verify the two upper bounds. First,

$$\begin{aligned} 2c_F a_\Psi &= 288L \frac{\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \lambda (L - (1-\theta)\mu) \\ &\leq 144L^2\gamma^2 \frac{\rho^2(1+\rho^2)}{(1+2\rho^2)(1-\rho^2)} \leq \frac{144}{64} \cdot \frac{33}{31}\rho^2 < \frac{12}{5}\rho^2 \leq \frac{3}{5}\mu\gamma = \frac{6}{5}\theta^2. \end{aligned}$$

Combining this with $\delta \geq 119/160$ and $\theta \leq 1/4$ gives

$$2c_F a_\Psi \leq \frac{6}{5}\theta^2 \leq \frac{3}{10}\theta \leq \theta\delta,$$

which proves $u \leq \theta/(2c_F)$. Second,

$$\begin{aligned} c_\Delta a_\Psi &= 72L^2 \frac{\rho^4(1+\rho^2)\gamma^2}{(1-\rho^2)^3} \lambda(L-(1-\theta)\mu) \\ &\leq 36L^3 \gamma^2 \frac{\rho^2(1+\rho^2)}{(1+2\rho^2)(1-\rho^2)} \leq \frac{36}{64} \cdot \frac{33}{31} L\rho^2 < \frac{3}{5} L\rho^2 \leq \frac{3}{20} \mu\gamma L \leq \frac{3\mu}{160}. \end{aligned}$$

On the other hand,

$$b_\Delta \delta = \frac{(1-\theta)\mu}{2} \delta \geq \frac{3\mu}{8} \cdot \frac{119}{160} > \frac{3\mu}{160}.$$

Hence $c_\Delta a_\Psi \leq b_\Delta \delta$, that is, $u \leq b_\Delta/c_\Delta$. Thus u is feasible.

Now apply Theorem A.1. Since $u = a_\Psi/\delta$,

$$\frac{2}{\theta} u c_F \tilde{G} = \frac{2c_F a_\Psi}{\theta \delta} \tilde{G} \leq \frac{6}{5} \frac{\theta^2}{\delta} \tilde{G} \leq \frac{192}{119} \theta \tilde{G}.$$

For the stochastic term carried by C , using $u \leq \theta/(2c_F)$ yields

$$\frac{2}{\theta} u \frac{24\rho^4(1+\rho^2)\gamma^2(n-1)\sigma^2}{(1-\rho^2)^3 R} \leq \frac{n-1}{6nLR} \sigma^2.$$

For the term D ,

$$\begin{aligned} \frac{2}{\theta} D &= \frac{\sigma^2}{nR} \frac{\mu\gamma^2}{2\theta} \left(\frac{\gamma L + 1}{(\theta + \mu\gamma)^2} \right) \\ &\leq \frac{\sigma^2}{nR} \frac{\mu\gamma^2}{2\theta} \frac{\gamma L + 1}{\theta^2} = \frac{\sigma^2}{nR} \frac{\gamma}{\theta} (\gamma L + 1) \leq \frac{9\sqrt{2}}{8} \frac{\sigma^2}{nR} \sqrt{\frac{\gamma}{\mu}}, \end{aligned}$$

where we used $\theta^2 = \mu\gamma/2$ and $\gamma L \leq 1/8$. It remains to control the initial term. Since the initialization is

$$x_i^{(0)} = y_i^{(0)} = z_i^{(0)} = 0,$$

we have $\Pi x^{(0)} = \Pi z^{(0)} = 0$, hence $\Psi_0 = 0$ and $\Phi_0 = F_0$. By L -smoothness of f and $\nabla f(x^*) = 0$,

$$f(x^{(0)}) - f(x^*) \leq \frac{L}{2} \|x^{(0)} - x^*\|^2 = \frac{L}{2} \Delta_x.$$

Also,

$$\frac{\theta^2}{2\gamma} + \frac{\mu\theta}{2} = \frac{\mu}{4} + \frac{\mu\theta}{2} \leq \frac{\mu}{4} + \frac{\mu}{8} \leq \frac{3L}{8},$$

because $\theta \leq 1/4$ and $\mu \leq L$. Therefore,

$$\Phi_0 = F_0 \leq \left(\frac{1}{2} + \frac{3}{8} \right) L\Delta_x = \frac{7}{8} L\Delta_x \leq L\Delta_x.$$

Substituting these estimates into Theorem A.1 proves (A.12). $\square$

Translation back to the paper notation. The detailed proof above follows the supplied file and is therefore written in terms of the auxiliary quantity $\tilde{G} = (nL)^{-1} \sum_i \|\nabla f_i(x^*)\|^2$. The paper itself reserves the notation G^* for the heterogeneity measure $f^* - \frac{1}{n} \sum_i f_i^*$. By Lemma A.1,

$$\tilde{G} \leq 2G^*.$$

Hence the entire proof can be restated in the paper notation without changing the argument structure; and in the homogeneous-optimum regime $G^* = 0$, the auxiliary term $\tilde{G}$ also vanishes, so the communication-complexity corollary stated in the main text follows exactly from the same proof chain.

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